

SOIL LOSS

Soil Loss Estimations

$$A = R \cdot K \cdot LS \cdot C \cdot P$$

where:

A= Annual soil loss due to erosion (t/ha/yr)

R= Rainfall erosivity factor

K= Soil erodibility factor

LS= topographic slope/length factor

P= Erosion Control Practice factor

C= Cover and management factor

Construction Area	Exposed Area (ha)	R	K	Equal Area Slope (%)	LS	P	C	A (t/ha/yr)	Soil Loss (t/yr)
Disturbed Catchment	2.25	5927	0.04	1.00	0.19	1.3	1	59	132

NOTE:

A CONSERVATIVE K FACTOR OF 0.04 HAS BEEN ADOPTED FOR THE PROJECT

P FACTOR VALUE OF 1.3 IS BASED ON AN INDUSTRY DEFAULT "CONSTRUCTION PHASE" IN ACCORDANCE WITH TABLE E11 (IECA 2025)

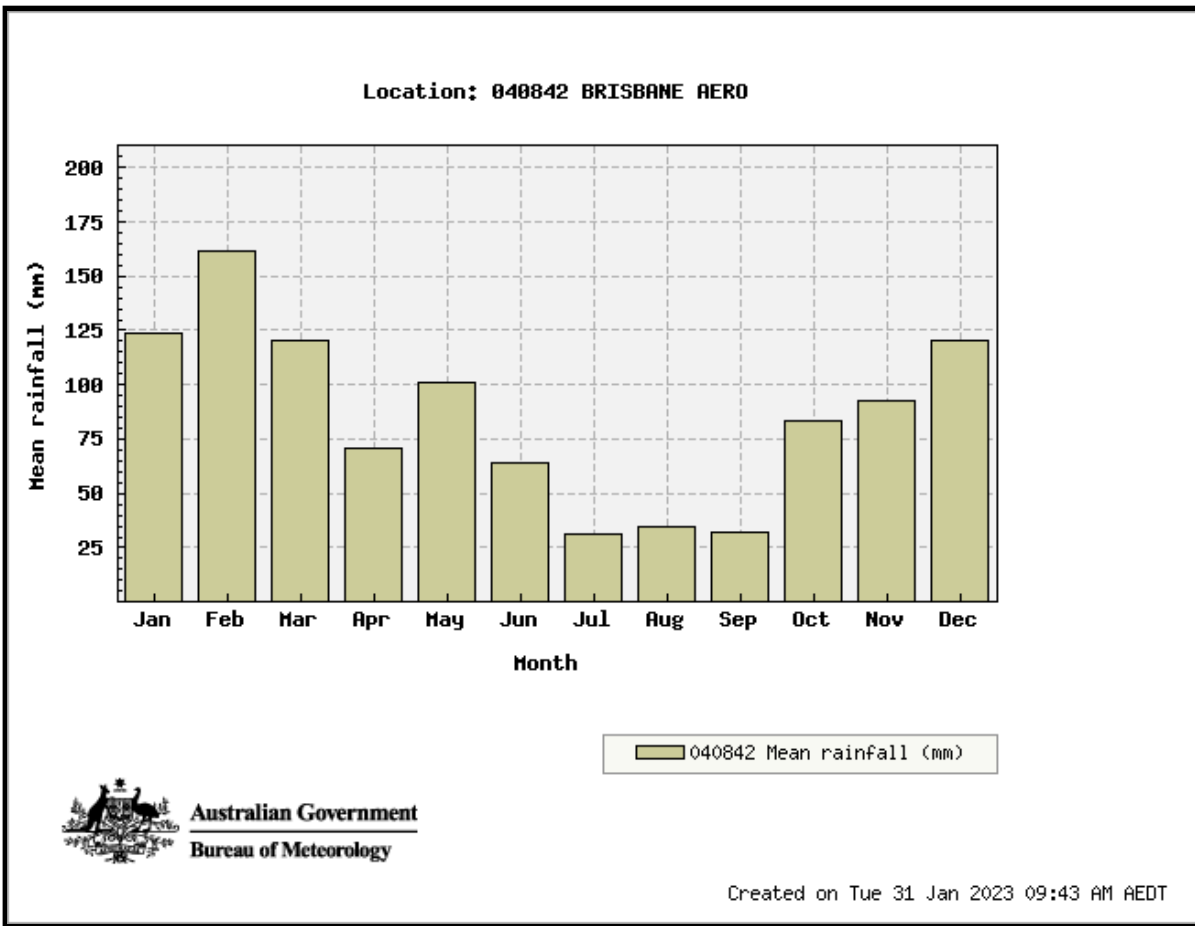
R FACTOR OF 5927 BRISBANE IS FROM TABLE E1 (IECA 2025)

SEDIMENT CONTROL

SEDIMENT CONTROL IS DETERMINED USING TABLE 4.5.1 (FROM IECA, 2025) THE TABLE INDICATES THAT THE SEDIMENT CONTROL STANDARD IS BASED ON THE CATCHMENT AREA AND SOIL LOSS. BASED ON THE CATCHMENT AREA TYPE 2 & 3 MEASURES ARE REQUIRED FOR THE SITE

Area limit (m ²) ^[1]	Soil loss rate limit (t/ha/yr) ^[2]		
	Type 1	Type 2	Type 3
250	N/A	N/A	[4]
1000	N/A	N/A	All cases
2500	N/A	> 75	75
>2500	> 150	150	75
>10,000	> 60	N/A	60

BRISBANE AERO MEAN RAINFALL



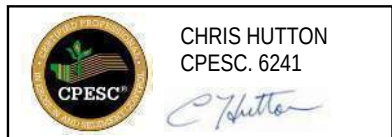
RAINFALL RISK ASSESSMENT

Erosion Risk Rating												Based on
Ave Monthly Rainfall (Source Table 4.4.2 IECA 2008)												
Monthly Data	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sept.	Oct.	Nov.	Dec.
Mean Rainfall	123.80	161.6	120.2	71.2	100.90	64.2	31.2	34.4	32.2	83.3	92.7	120.2
Erosion Risk	High	High	High	Moderate	High	Moderate	Low	Low	Low	Moderate	Moderate	High
Very Low Risk: 0 to 30mm												
Low Risk: 30+ to 45mm												
Moderate Risk: 45+ to 100mm												
High Risk: 100+ to 225mm												
Extreme Risk: >225mm												

NOTE: RAINFALL RISK IS MODERATE TO LOW BETWEEN JUNE TO NOVEMBER. WHERE POSSIBLE IT IS RECOMMENDED THAT BULK EARTHWORKS BE UNDERTAKEN DURING LOWER RISK MONTHS. UP TO DATE WEATHER FORECASTS SHOULD BE MONITORED PRIOR TO UNDERTAKING SOIL DISTURBANCE ACTIVITIES. RAINFALL DATA FROM BRISBANE AERO RAINFALL STATION NO.040842

EROSION RISK RATING

Applicable Area ID	Erosion Risk Rating	Soil Loss Rate (t/ha/year)	Advance Land Clearing Allowed (weeks work)	Max days to Stabilisation	Staged Construction and Stabilisation of Earth Batters >6H:1V	Stockpiles stabilised
Disturbed Catchment	Very Low	0 to 150	8	30 (60%)		
	Low	150 to 225	8	30 (70%)		
	Moderate	225 to 500	6	20 (70%)	✓	
	High	500 to 1500	4	10 (75%)	✓	✓
	Extreme	> 1500	2	5 (80%)	✓	✓



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APPROVED

DATE

26-05-2026

CLIENT

NORTHROP CONSULTING ENGINEERS

PROJECT

SKYGATE PLAY

LOCATION

LOT SKG26, THE CIRCUIT, BRISBANE AIRPORT

DRAWING TITLE

EROSION AND SEDIMENT CONTROL - RISK ASSESSMENT

JOB No.

ESC-0606

DRAWING No. / REVISION

DR-02 / a